Anatomy of IoT Cyber Attacks

Smart Sensors and IoT (20MSS82)

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Introduction

Why IoT Security Matters

- IoT topology spans **hardware, networking, protocols, cloud, and OS**
- Diverse attack surface creates unique vulnerabilities
- Understanding threats helps architects design preventative measures

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Three Major IoT Attack Categories:

1. **Mirai** - Massive DDoS via compromised IoT devices
2. **Stuxnet** - Nation-state cyber weapon targeting SCADA
3. **Chain Reaction** - PAN mesh network exploitation

Mirai Botnet Attack

Overview

- **Date:** August 2016
- **Type:** Malware infecting Linux IoT devices
- **Result:** Largest DDoS attack in history

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High-Profile Targets:

- Krebs on Security (623 Gbps attack)
- Dyn DNS Provider
- Lonestar Cell (Liberia)

Collateral Damage: Amazon, GitHub, Netflix, PayPal, Reddit, Twitter

Mirai Attack Methodology

Phase 1: Scan for Victims

- Asynchronous TCP SYN scan on random IPv4 addresses
- Target ports: **SSH/Telnet (23, 2323)**
- Scan rate: ~250 bytes/sec (constrained by IoT hardware)
- Blacklisted 3.4M IPs (US Postal, HP, GE, DoD)

Mirai Attack Methodology

Phase 2: Brute Force Telnet

- Attempted login with **62 username/password pairs**
- Dictionary attack with 10 random pairs per attempt
- Successful logins reported to **C2 server**
- Later variants: Remote Code Execution (RCE) exploits

Mirai Attack Methodology

Phase 3: Infect

- Loader identifies OS and installs device-specific malware
- Kills competing processes on ports 22/23
- **Non-persistent:** Does not survive reboot
- Process name obfuscated to hide presence
- Bot stays dormant until attack command received

Targeted Devices

Device Types:

- IP Cameras
- DVRs
- Consumer Routers
- VoIP Phones
- Printers
- Set-top Boxes

Architectures: 32-bit ARM, MIPS, x86

Mirai Infection Timeline

Time	Event
Aug 1, 2016	First scan from US hosting site
+120 min	First successful infection
+1 min	834 devices infected
+20 hours	64,500 devices infected

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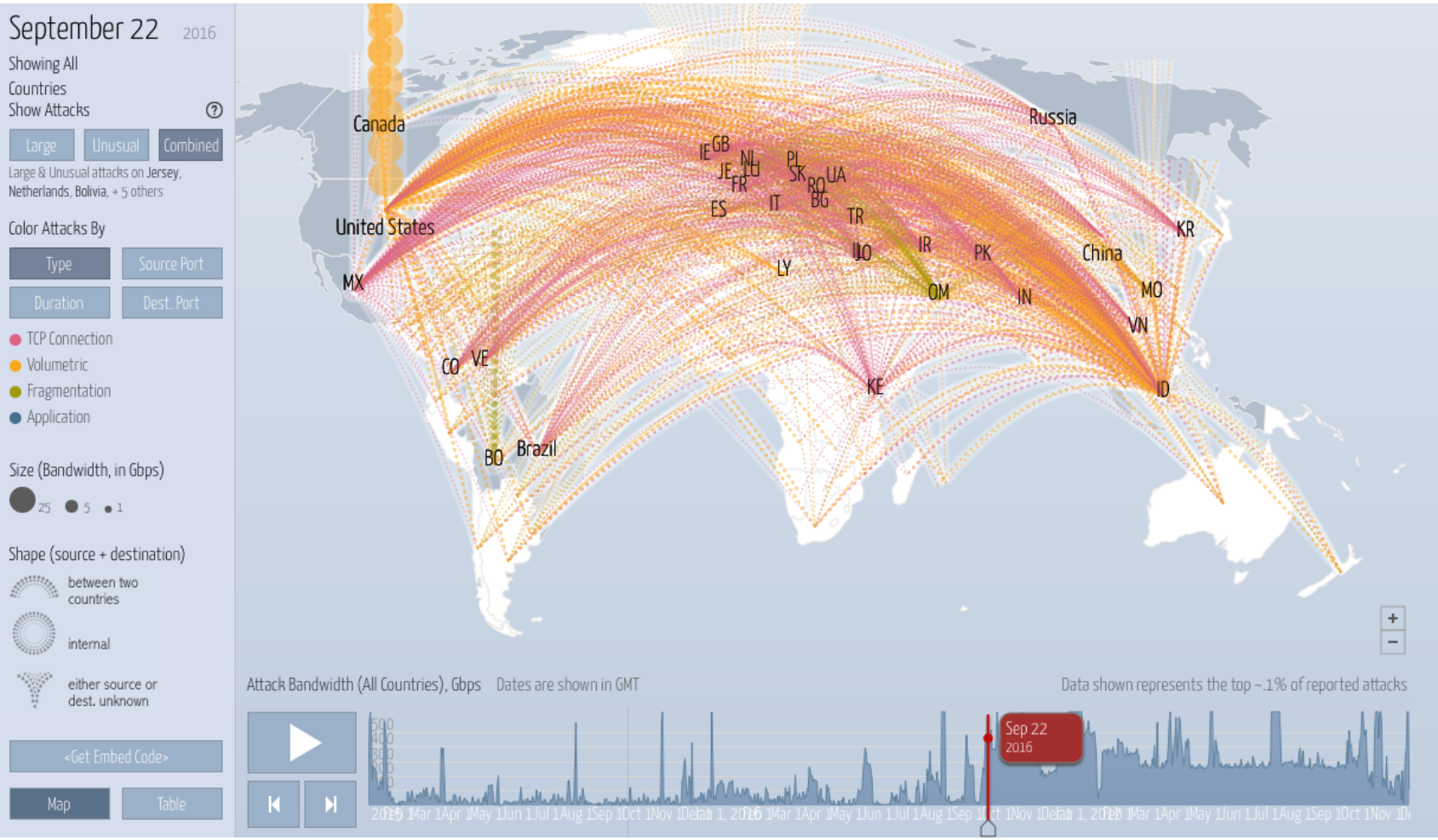
Doubling time: 75 minutes

Total infected: 600,000 IoT devices

Mirai Global Attack Visualization



Mirai DDoS Attack - September 22, 2016 (digitalattackmap.com)

Mirai DDoS Attack Types

- **SYN Floods**
- **GRE IP Network Floods**
- **STOMP Floods**
- **DNS Floods**

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Statistics (5 months):

- 15,194 attack commands issued
- 5,042 internet sites targeted
- Top botnet locations: Brazil (15%), Colombia (14%), Vietnam (12.5%)

Stuxnet

Overview

- **First documented nation-state cyber weapon**
- **Target:** Iranian nuclear enrichment facility (Natanz)
- **Method:** Worm targeting Siemens PLCs via SCADA
- **Damage:** 1,000+ uranium centrifuges destroyed

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Sophistication: Used **4 zero-day exploits simultaneously**

Stuxnet Attack Flow

Phase 1: Initial Infection

- Spread via **USB drive** insertion
- Exploited Windows vulnerabilities
- Installed **rootkit** (user-mode + kernel-mode)
- Used **stolen signed driver** from Realtek
- Bypassed antivirus detection

Stuxnet Attack Flow

Phase 2: Windows Attack & Spread

- Searched for **Siemens WinCC/Step-7** SCADA software
- Connected to C2 servers for payload updates
- Targeted `s7otbdx.dll` (PLC communication library)
- Exploited **hardcoded password** (another zero-day)
- Acted as **Man-in-the-Middle** between WinCC and PLC
- Recorded normal centrifuge operation

Stuxnet Attack Flow

Phase 3: Destruction

- **Replayed pre-recorded data** to SCADA (no alarms)
- Manipulated PLCs controlling centrifuge motors
- Targeted specific frequencies: **807 Hz and 1210 Hz**

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Attack Pattern:

- 15-50 minute damage increments
- Separated by 27 days of normal operation
- Resulted in cracked rotors and improper uranium enrichment

Chain Reaction

Overview

- **Type:** Academic research / proof-of-concept
- **Target:** Philips Hue smart light bulbs
- **Protocol:** Zigbee Light Link (ZLL)
- **Key insight:** No internet connection required

Philips Hue Bulb

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Philips Hue Bulb

Implications: Scalable to smart city attacks

Zigbee Vulnerabilities

Protocol Weaknesses:

- ZLL messages **not encrypted or signed**
- Master key shared among ZLL alliance (leaked)
- Proximity check for network joining
- OTA firmware updates encrypted but breakable

Chain Reaction Attack Phases

Phase 1 & 2: Break Encryption

- Used **Correlation Power Analysis (CPA)**
- Used **Differential Power Analysis (DPA)**
- Cracked AES-CCM encryption keys
- Broke firmware signing

Chain Reaction Attack Phases

Phase 3: Bypass Proximity Check

- Found **zero-day** in Atmel AtMega bootloader
- Proximity check only valid on scan request
- Starting with different message bypasses check
- Enabled joining any Zigbee network remotely

Chain Reaction Attack Phases

Phase 4: Worm Propagation

- Infected bulb joins network using stolen master key
- Sends malicious payload to neighboring lights
- Spreads via **Percolation Theory**
- Can disable firmware update capability permanently

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Demo: Researchers flew drone to hijack campus lights

Key Security Concepts

Definitions

Term	Description
Root of Trust	Immutable trusted boot source (ROM)
Secure Boot	Verified signature chain from RoT to OS
TEE	Trusted Execution Environment on processor
Stack Canaries	Guards against stack overflow attacks

Cryptographic Fundamentals

- **Private Key:** Never released, stored securely, encrypts hashes
- **Public Key:** Distributed for signature verification
- **AES-CCM:** Encryption standard for Zigbee (IEEE 802.15.4)

Lessons Learned

Prevention Strategies

1. **Change default credentials** on all IoT devices
2. **Disable unnecessary ports** (Telnet, SSH)
3. **Implement secure boot** with verified signatures
4. **Regular firmware updates** with signed packages
5. **Network segmentation** for IoT devices
6. **Monitor for anomalous traffic** patterns

Summary

Attack	Vector	Impact
Mirai	Default passwords, Telnet	623 Gbps DDoS
Stuxnet	USB, Zero-days, SCADA	Physical destruction
Chain Reaction	Zigbee, Power analysis	Mesh takeover

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Key Takeaway: IoT security requires defense-in-depth across all layers

Thank You

Questions?

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Course: Smart Sensors and IoT (20MSS25)